Respiratory Critical Care – Chapter Summary 1/2

Physiology • Monitoring • Respiratory Support

ICU / EDIC Board Review • MCQ-Oriented

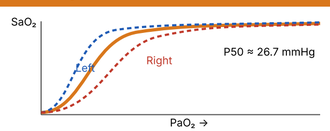
1. Lung Mechanics

- **Equation of motion:** Paw = (V/C) + (R × Flow) + $PEEP_{total}$
- **Compliance:** normal static 60–100 mL/cmH_2O; ARDS often < 40
- **Resistance:** normal 5–10 cmH_2O/L/s in intubated (↑ with small ETT, bronchospasm)
- **Respiratory system C** = lung + chest wall (in series): $1/C_{rs} = 1/C_L + 1/C_{cw}$
- ↓C_{cw}: obesity, abdominal hypertension, burns eschar, kyphoscoliosis
- **FRC** ↓ supine, obesity, anesthesia, ARDS; ↑ in COPD (hyperinflation)
- **Closing capacity** > FRC (elderly, supine) → airway closure → V/Q mismatch
- **Surfactant** (type II pneumocytes) ↓ surface tension → alveolar stability

2. Control of Breathing

- **Central chemoreceptors** (medulla): respond to CSF H^+ ← $PaCO_2$ – main minute-to-minute drive
- **Peripheral** (carotid > aortic bodies): PaO_2 < 60, ↓pH, ↑$PaCO_2$
- **$PaCO_2$ = 0.863 × VCO_2 / VA** → CO_2 depends on alveolar ventilation
- Opioids, sedatives, hypothyroid, chronic hypercapnia ↓ CO_2 response slope
- **P0.1** (airway occlusion pressure): > 3.5–4 cmH_2O = high drive (risk P-SILI); < 1 = low drive
- **Cheyne–Stokes:** HF, stroke (unstable loop gain)
- O_2 in COPD → ↑$PaCO_2$ mainly via **↑V/Q mismatch (loss of HPV) + Haldane effect**, not loss of hypoxic drive

3. Oxygen Transport



- **Right shift (↓affinity):** ↑CO_2, ↑H^+, ↑temp, ↑2,3-DPG
- **Left shift:** CO, MetHb, alkalosis, hypothermia, fetal Hb, stored blood
- **CaO_2** = 1.34 × Hb × SaO_2 + 0.003 × PaO_2 (≈ 20 mL/dL)
- **DO_2** ≈ 1000 mL/min; **VO_2** ≈ 250 mL/min; **O_2ER** ≈ 25%
- **SvO_2** ≈ 65–70%; $ScvO_2$ normally 2–5% higher; ↓ = ↓DO_2 or ↑VO_2; ↑ = sepsis/shunting, cyanide

4. CO_2 & Dead Space

- CO_2 carried: **HCO_3^- ~70%**, carbamino ~20%, dissolved ~10%
- **Haldane effect:** oxygenation of Hb releases CO_2
- **Bohr effect:** ↑CO_2/H^+ releases O_2
- **VD/VT** = ($PaCO_2$ − $PECO_2$)/$PaCO_2$; normal 0.2–0.35; ARDS > 0.6 = ↑mortality
- **Ventilatory ratio** = (VE × $PaCO_2$)/(PBW × 100 × 37.5); > 1 = ↑dead space
- **Causes ↑dead space:** PE, low CO, high PEEP/overdistension, COPD, apparatus (HME)

5. Approach to Hypoxemia

Mechanism	A-a	Response to O_2
↓ PiO_2 (altitude)	Normal	Good
Hypoventilation	Normal	Good
V/Q mismatch	↑	Good
Diffusion limit	↑	Good
Shunt	↑	**Poor (> 30%)**

- **PAO_2** = FiO_2 (Patm − 47) − $PaCO_2$/0.8
- Expected A-a ≈ Age/4 + 4 (room air)
- Intracardiac shunt (PFO with ↑RAP), hepatopulmonary syndrome (orthodeoxia), AVM
- **Low SvO_2** worsens hypoxemia in presence of shunt

6. Acid–Base Essentials

Disorder	Expected compensation
Met acidosis	$PaCO_2$ = 1.5 × HCO_3 + 8 ± 2
Met alkalosis	$PaCO_2$ ↑ 0.7 per 1 ↑HCO_3
Acute resp acid.	HCO_3 ↑1 / 10 $PaCO_2$
Chronic resp acid.	HCO_3 ↑3.5 / 10 $PaCO_2$
Acute resp alk.	HCO_3 ↓2 / 10 $PaCO_2$
Chronic resp alk.	HCO_3 ↓5 / 10 $PaCO_2$

- **AG** = Na − (Cl + HCO_3); corrected +2.5 per 1 g/dL ↓albumin
- **Δ-ratio** (ΔAG/ΔHCO_3): < 1 + NAGMA; 1–2 pure HAGMA; > 2 + met alk
- **Osmolar gap** > 10 → toxic alcohols
- **Stewart:** SIDa normal ≈ 40; ↓SID (hyperchloremia, NaCl 0.9%) → acidosis
- Post-hypercapnic met alkalosis after rapid ventilation of chronic CO_2 retainers

7. Respiratory Monitoring

Pulse oximetry pitfalls

- **CO-Hb:** falsely normal SpO_2 → co-oximetry
- **MetHb:** SpO_2 ≈ 85% regardless of PaO_2 ("saturation gap")
- Poor perfusion, motion, nail polish; **overestimates in dark skin**

Capnography

- Pa-$EtCO_2$ gap normally 2–5 → ↑ with dead space
- Loss of waveform: disconnection / esophageal / dislodged tube
- Shark-fin: bronchospasm; CPR $EtCO_2$ < 10 = poor CPR; abrupt ↑ = ROSC

Esophageal pressure (Pes)

- Transpulmonary P_L = Paw − Pes
- End-inspiratory P_L < 20–25 (stress); end-expiratory P_L ≥ 0 to prevent collapse (obesity)
- **Pmus / ΔPocc** estimate patient effort

8. Lung Ultrasound – BLUE Protocol

Profile	Diagnosis
A-profile + DVT	Pulmonary embolism
A-profile, no DVT, + PLAPS	Pneumonia
A-profile, nothing else	COPD / asthma
B-profile (bilateral B-lines + sliding)	Pulmonary edema
B'-profile (B-lines, no sliding)	Pneumonia
A'-profile + lung point	Pneumothorax

- Lung point = 100% specific for pneumothorax
- Diaphragm thickening fraction < 20–30% → weaning failure risk
- Lung ultrasound score (LUS) tracks aeration / recruitment

9. Oxygen Therapy & HFNC

Device	Flow	Approx. FiO_2
Nasal cannula	1–6 L/min	24–44%
Simple face mask	5–10 L/min	35–55%
Venturi	Set	24–60% (fixed)
Non-rebreather	10–15 L/min	60–90%
HFNC	30–60 L/min	21–100%

HFNC benefits: washout of dead space, small PEEP (~1 cmH_2O / 10 L), precise FiO_2, humidification, ↓work of breathing

- **FLORALI:** HFNC ↓ 90-day mortality vs NIV in hypoxemic ARF (P/F ≤ 300)
- **ROX** = (SpO_2/FiO_2)/RR: ≥ 4.88 success; < 3.85 failure at 12 h
- **Targets:** SpO_2 92–96% general; 88–92% hypercapnic risk
- Delayed intubation after HFNC failure → ↑mortality

10. Non-Invasive Ventilation

✔ Strong evidence: AECOPD with pH < 7.35 & $PaCO_2$ > 45; cardiogenic pulmonary edema (CPAP or BiPAP); immunocompromised; post-extubation in high-risk / hypercapnic; pre-oxygenation

✘ Contraindications: arrest / imminent arrest; coma or unable to protect airway; shock; uncontrolled vomiting / UGIB; facial trauma or surgery; undrained pneumothorax; copious secretions

- **Start:** IPAP 10–12, EPAP 4–5 → titrate IPAP for VT 6–8 mL/kg & ↓$PaCO_2$; EPAP for oxygenation
- Reassess ABG at **1–2 h**; no improvement → intubate
- **HACOR score** > 5 at 1 h predicts NIV failure in hypoxemia
- Helmet NIV: better tolerated, allows higher PEEP

11. Invasive Ventilation – Modes & Setup

Mode	Set	Variable
VC-AC	VT, RR, flow	Pressures
PC-AC	Pinsp, Ti, RR	VT
PRVC	Target VT	Pressure adapts
PSV	PS, PEEP	VT, RR, Ti
SIMV	Mandatory + PS	Mixed
APRV	P-high, T-high, P-low, T-low	VT
NAVA / PAV+	Gain	Proportional to effort

Initial settings: VT 6–8 mL/kg PBW • RR 14–20 • PEEP 5+ • FiO_2 to SpO_2 92–96% • I:E 1:2

- **PBW** male = 50 + 0.91 (height cm − 152.4); female = 45.5 + 0.91 (height cm − 152.4)
- ↑ Oxygenation: ↑FiO_2, ↑PEEP, ↑mean airway pressure
- ↑ CO_2 clearance: ↑RR or VT, ↓dead space (remove HME)

12. Ventilator Troubleshooting



- **↑Ppeak, normal Pplat** → ↑resistance: secretions, bronchospasm, biting, kinked / small ETT
- **↑Ppeak and ↑Pplat** → ↓compliance: pneumothorax, mainstem intubation, atelectasis, edema, ARDS, abdominal hypertension, auto-PEEP

Acute deterioration – "DOPES": Displacement, Obstruction, Pneumothorax, Equipment failure, Stacked breaths (auto-PEEP) → disconnect & bag with 100% O_2

- **Auto-PEEP:** expiratory flow not reaching zero; expiratory hold; ↓RR, ↓VT, ↑expiratory time, bronchodilators
- **Dyssynchrony:** ineffective, double, auto- & reverse triggering, flow starvation, premature / delayed cycling

13. Lung Protection & VILI

Targets: VT 6 mL/kg PBW (4–8) • Pplat ≤ 30 • ΔP (Pplat − PEEP) ≤ 15 • pH ≥ 7.20–7.25

- **ARDSNet (ARMA):** 6 vs 12 mL/kg → ↓ mortality (31% vs 40%)
- **VILI:** barotrauma, volutrauma, atelectrauma, biotrauma, ergotrauma
- **Stress** = P_L; **strain** = VT/FRC (> 2 injurious)
- **Mechanical power** (energy/min) > ~17 J/min associated with mortality
- **P-SILI:** vigorous spontaneous efforts → large P_L swings, pendelluft
- Protective VT also matters in non-ARDS patients (aim 6–8)
- **Recruitment maneuvers:** not routine (ART: ↑mortality)

14. Liberation & Tracheostomy

- **Readiness:** cause improving, FiO_2 ≤ 0.4–0.5, PEEP ≤ 5–8, P/F > 150–200, minimal vasopressors, follows commands
- **SBT:** PS 5–8 / PEEP 0–5 or T-piece, 30–120 min
- **Fail if:** RR > 35, SpO_2 < 90%, HR > 140 or ↑20%, SBP > 180 or < 90, anxiety, diaphoresis, paradoxical breathing
- **RSBI** = RR/VT (L) < 105
- **Daily SAT + SBT** (ABC trial) ↓ ventilator days & mortality
- **Post-extubation:** HFNC (low risk); NIV ± HFNC (high risk, hypercapnic, obese)
- **Cuff-leak** < 110 mL or < 12–24% → methylprednisolone ≥ 4 h before
- **Weaning failure causes:** cardiac (weaning-induced pulmonary edema), weakness, ↑load, delirium, metabolic
- **Tracheostomy:** early (≤ day 4) vs late (≥ day 10) – no mortality benefit (TracMan)

15. Airway Management

- **Physiologically difficult airway:** hypoxemia, hypotension, severe metabolic acidosis, RV failure
- **MACOCHA** ≥ 3 predicts difficult ICU intubation
- **Pre-oxygenation:** NIV better than face mask (PREOXI); HFNC for apneic oxygenation
- **Video laryngoscopy** ↑ first-pass success (DEVICE)
- **Induction:** ketamine / etomidate – reduce dose in shock; push-dose vasopressor ready
- **Rocuronium** 1.2 mg/kg ≈ succinylcholine 1.5 mg/kg
- **Sux CI:** hyperkalemia, burns / crush > 24–48 h, denervation, myopathy, MH
- Confirm with **waveform capnography**

CICO → scalpel–bougie–tube cricothyroidotomy

16. High-Yield MCQ Pearls

- ☑ Hypoventilation & altitude: normal A-a gradient
- ☑ Shunt > 30%: PaO_2 barely rises with FiO_2
- ☑ CO poisoning: normal SpO_2 – use co-oximetry
- ☑ MetHb: SpO_2 fixed ≈ 85%
- ☑ ↑Ppeak + normal Pplat = resistance problem
- ☑ Auto-PEEP + hypotension → disconnect
- ☑ ΔP is the strongest predictor of ARDS survival
- ☑ HFNC beat NIV in hypoxemic ARF (FLORALI)
- ☑ P0.1 > 4 cmH_2O: excessive drive
- ☑ Lung point = pneumothorax (100% specific)
- ☑ Pair SAT + SBT daily
- ☑ Early tracheostomy: no mortality benefit

Respiratory Critical Care – Chapter Summary 2/2 Diseases • Emergencies • Rescue Therapies

ICU / EDIC Board Review • MCQ-Oriented

1. ARDS – Definition & Pathology

Global definition (2023): onset ≤ 1 week; bilateral opacities (CXR / CT / ultrasound); not fully explained by cardiac failure / overload

Category	Oxygenation criterion
Intubated	P/F ≤ 300 or S/F ≤ 315, PEEP ≥ 5 Mild 200–300 • Mod 100–200 • Severe ≤ 100
Non-intubated	HFNC ≥ 30 L/min or NIV/CPAP ≥ 5; P/F ≤ 300 or S/F ≤ 315
Resource-limited	S/F ≤ 315 (no PEEP / flow requirement)

- **Direct:** pneumonia, aspiration, contusion, inhalation, drowning
- **Indirect:** sepsis (most common), pancreatitis, trauma, transfusion, burns
- **Pathology:** diffuse alveolar damage – exudative (hyaline membranes) → proliferative → fibrotic
- S/F valid only when SpO_2 ≤ 97%

2. ARDS – Management Ladder



- **Treat cause**, conservative fluids after shock resolves (FACTT)
- **VT 6 mL/kg PBW**, Pplat ≤ 30, ΔP ≤ 15 (ARMA)
- **Higher PEEP** in moderate–severe (PEEP/FiO_2 table, P_L-guided or best compliance)
- **Prone ≥ 16 h/day** if P/F < 150 on FiO_2 ≥ 0.6 – ↓mortality (PROSEVA)
- **NMB** 48 h: no routine benefit (ROSE); use for severe dyssynchrony / refractory hypoxemia
- **Inhaled vasodilators:** improve oxygenation, no mortality benefit (rescue only)
- **Steroids:** dexamethasone early in moderate–severe (DEXA-ARDS); COVID & severe CAP benefit
- **Avoid:** HFOV (OSCILLATE), routine recruitment (ART), β-agonists

3. ECMO

	VV-ECMO	VA-ECMO
Supports	Gas exchange	Heart + lungs
Indication	Refractory ARDS, bridge to transplant	Cardiogenic shock, ECPR, massive PE
Cannulas	Femoral → IJ	Femoral vein → femoral artery

EOLIA criteria: P/F < 50 for > 3 h, or < 80 for > 6 h, or pH < 7.25 with $PaCO_2$ ≥ 60 for > 6 h (despite optimal care)

- **Relative CI:** MV > 7 days at high settings, irreversible disease, contraindication to anticoagulation, severe brain injury
- **Ultra-protective** ventilation on ECMO (rest settings, Pplat ≤ 25)
- **Oxygenation** ↑ with blood flow; **CO_2 removal** ↑ with sweep gas flow
- **Harlequin (north–south) syndrome:** peripheral VA + poor lung function → hypoxic upper body (monitor right radial ABG)
- VV recirculation ↓ efficiency; $ECCO_2R$ alone: no benefit (REST)

4. Life-Threatening Asthma

Life-threatening: SpO_2 < 92%, PEF < 33%, silent chest, cyanosis, exhaustion, confusion, arrhythmia, hypotension, **"normal" $PaCO_2$**
Near-fatal: ↑$PaCO_2$ or need for ventilation

- **Rx:** O_2 to SpO_2 93–95%, nebulized salbutamol (back-to-back) + ipratropium, hydrocortisone 100 mg IV / prednisolone 40–50 mg, **$MgSO_4$ 2 g IV over 20 min**
- **Refractory:** IV salbutamol / adrenaline, ketamine, heliox, inhaled anesthetics
- **Salbutamol toxicity:** lactic acidosis, hypokalemia, tachycardia
- **Intubate with ketamine**, large ETT
- **Ventilate:** low RR (8–12), VT 6–8, high inspiratory flow, I:E ≥ 1:4, permissive hypercapnia, Pplat < 30, minimal external PEEP
- Peri-intubation arrest → disconnect, compress chest, exclude tension pneumothorax

5. AECOPD

- **Triggers:** viral > bacterial infection, pollution, PE, HF, non-adherence
- **Controlled O_2:** SpO_2 88–92% (Venturi)
- **Bronchodilators:** SABA ± SAMA nebulized
- **Steroids:** prednisolone 40 mg × 5 days (REDUCE)
- **Antibiotics** if ↑sputum purulence + ↑volume or ↑dyspnea (Anthonisen) or ventilated

NIV: pH < 7.35 & $PaCO_2$ > 45 → ↓intubation & mortality; pH < 7.25 → NIV with low threshold for intubation

- **Ventilation:** long expiratory time, monitor auto-PEEP, target baseline $PaCO_2$ (avoid post-hypercapnic alkalosis)
- **Mimics:** PE, pneumothorax, HF, pneumonia

6. Severe Community-Acquired Pneumonia

ATS/IDSA severe CAP: 1 major (septic shock on vasopressors or MV) or ≥ 3 minor: RR ≥ 30, P/F ≤ 250, multilobar, confusion, BUN ≥ 20, WBC < 4000, plt < 100k, T < 36 °C, hypotension needing fluids

- **Empiric:** β-lactam + macrolide (or + respiratory FQ)
- Add **MRSA** / **Pseudomonas** cover only if risk factors (prior isolation, recent hospitalization + IV antibiotics)
- **Hydrocortisone** 200 mg/day in severe CAP → ↓mortality (CAPE COD)
- **Duration:** ≥ 5 days & clinically stable

Organism	Clue
Legionella	Hyponatremia, diarrhea, ↑LFT, urine antigen
S. aureus (PVL)	Post-influenza, necrotizing, hemoptysis
Klebsiella	Alcohol, upper lobe, cavitation
Pneumococcus	Most common; urine antigen

7. HAP / VAP

- **HAP:** ≥ 48 h after admission; **VAP:** ≥ 48 h after intubation
- **Diagnosis:** new infiltrate + fever, purulent secretions, ↑WBC, ↓oxygenation; quantitative ETA / BAL
- **Empiric:** anti-pseudomonal β-lactam; add MRSA cover (vancomycin / linezolid) if risk
- **MDR risk:** IV antibiotics in prior 90 days, septic shock, ARDS before VAP, ≥ 5 days hospital, acute RRT
- Double Gram-negative cover if MDR risk + high local resistance
- **Duration:** 7 days (de-escalate by culture)

Prevention bundle: head-up 30–45°, daily SAT/SBT, subglottic secretion drainage, minimize sedation, early mobility, cuff pressure 20–30 cmH_2O

8. Viral & Opportunistic Infections

- **Influenza:** oseltamivir for all severe cases, even > 48 h; watch S. aureus superinfection, aspergillosis
- **COVID-19:** dexamethasone 6 mg × 10 d if O_2-requiring; + IL-6 inhibitor or baricitinib if progressing; awake prone positioning
- **PCP:** CD4 < 200, ↑LDH, ↑β-D-glucan, diffuse GGO, pneumothorax; TMP-SMX + steroids if PaO_2 < 70 or A-a ≥ 35
- **Invasive aspergillosis:** neutropenia, halo / air-crescent sign, BAL galactomannan → voriconazole
- **Mucormycosis:** diabetic ketoacidosis, reverse halo → liposomal amphotericin + debridement
- **CMV:** transplant → ganciclovir
- **Nocardia:** cavities + brain abscess → TMP-SMX
- **TB:** airborne isolation; RIPE (INH neuropathy – give pyridoxine; ethambutol optic neuritis; rifampicin interactions)

9. Pulmonary Embolism

Risk	Features	Management
High	SBP < 90 ≥ 15 min, vasopressors, arrest	Systemic thrombolysis (alteplase 100 mg / 2 h; arrest 50 mg bolus); embolectomy / CDT / VA-ECMO if contraindicated
Int-high	RV dysfunction + ↑troponin	Anticoagulation, monitor; rescue reperfusion if deteriorates (PEITHO: ↑bleeding)
Int-low	sPESI ≥ 1, one marker	Anticoagulation
Low	sPESI 0	DOAC, early discharge

- **Pre-test:** Wells / Geneva; PERC (low risk); age-adjusted D-dimer (age × 10 µg/L)
- **Imaging:** CTPA; V/Q in pregnancy / renal failure; unstable → bedside echo (RV dilatation, McConnell sign)
- **RV failure:** small fluid boluses only, norepinephrine, avoid intubation if possible (collapse)
- **Anticoagulation:** UFH if thrombolysis likely / renal failure; otherwise LMWH / DOAC

10. Pulmonary Hypertension & RV Failure

- **PH (2022):** mPAP > 20 mmHg on RHC
- **Pre-capillary:** PAWP ≤ 15 + PVR > 2 WU; **post-capillary:** PAWP > 15
- **Groups:** 1 PAH • 2 left heart disease (most common) • 3 lung disease / hypoxia • 4 CTEPH • 5 unclear / multifactorial
- **PAH drugs:** PDE5i, ERA, riociguat, prostacyclin analogues

ICU RV failure:
• Preload: avoid overload, diuresis / CRRT if congested
• Afterload: correct hypoxia, hypercapnia, acidosis; iNO / inhaled epoprostenol
• Contractility: dobutamine, milrinone, levosimendan
• Perfusion: keep MAP > PAP (norepinephrine, vasopressin)
• Sinus rhythm; avoid high PEEP / Pplat; intubation = high-risk event

11. ILD in the ICU

Acute exacerbation of IPF: new bilateral GGO within 1 month, not explained by HF / overload; mortality > 50%; MV outcomes very poor → discuss goals / transplant candidacy

- **Rx (weak evidence):** high-dose steroids, broad antibiotics; cyclophosphamide harmful (EXAFIP)
- **Rapidly progressive ILD:** anti-MDA5 dermatomyositis, anti-synthetase syndrome, vasculitis
- **Drug-induced:** amiodarone, methotrexate, nitrofurantoin, checkpoint inhibitors, **bleomycin (avoid high FiO_2)**
- **Hypersensitivity pneumonitis:** exposure, mosaic attenuation, BAL lymphocytosis
- **Sarcoidosis:** non-caseating granulomas, hilar nodes, hypercalcemia
- **Organizing pneumonia:** migratory consolidations, reverse halo, steroid-responsive
- **AEP (acute eosinophilic pneumonia):** new smoker, BAL eosinophils > 25%, dramatic steroid response

12. Pleural Disease

Light's criteria (exudate if ANY)

- Pleural / serum protein > 0.5
- Pleural / serum LDH > 0.6
- Pleural LDH > 2/3 ULN serum LDH
- Diuresed HF misclassified as exudate → serum–pleural albumin gradient > 1.2 g/dL = transudate

Finding	Meaning
pH < 7.2 / glucose < 3.3 mmol/L	Complicated parapneumonic → drain
Frank pus	Empyema → drain
TG > 110 mg/dL	Chylothorax
Hct ratio > 0.5	Hemothorax
ADA > 40 U/L	TB
Amylase ↑	Pancreatitis, esophageal rupture

- **Empyema:** antibiotics + drain + intrapleural tPA 10 mg / DNase 5 mg BD × 3 d (MIST-2); VATS if fails
- **Re-expansion edema:** drain ≤ 1.5 L at once

13. Pneumothorax & Air Leak

Tension: clinical diagnosis – hypotension, ↑Ppeak/Pplat, desaturation, absent breath sounds → immediate decompression (4th–5th ICS anterior axillary line; or 2nd ICS MCL), then chest drain

- **Ventilated patients:** any pneumothorax → chest drain
- **US:** absent sliding, no B-lines, barcode sign (M-mode), lung point
- **Bronchopleural fistula:** persistent air leak; minimize Pplat / PEEP / VT / Ti, lowest suction; endobronchial valves or surgery; independent lung ventilation (severe)
- **Hemothorax:** large-bore drain; thoracotomy if > 1500 mL initially or > 200 mL/h for 2–4 h

14. Massive Hemoptysis & DAH

- **Massive:** causing airway compromise / hypoxemia (volume definitions 100–600 mL/24 h)
- **Source:** bronchial arteries ~90%
- **Causes:** bronchiectasis, TB, aspergilloma, malignancy, CF, vasculitis

Rx: bleeding side down • secure airway (ETT ≥ 8 for bronchoscopy; selective intubation of non-bleeding lung / bronchial blocker) • reverse coagulopathy • TXA • CT angiography → **bronchial artery embolization** • surgery if fails

- **DAH:** ↓Hb, diffuse infiltrates, progressively bloodier BAL, hemosiderin-laden macrophages, ↑DLCO
- **DAH causes:** GPA, MPA, anti-GBM, SLE, HSCT, coagulopathy, cocaine
- **DAH Rx:** pulse methylprednisolone + cyclophosphamide / rituximab; plasma exchange in anti-GBM

15. Other Respiratory Emergencies

- **Smoke inhalation:** early intubation if stridor, facial burns, soot, hoarseness
- **CO poisoning:** normal SpO_2; 100% O_2 (t½ 320 → 74 min); HBO if LOC, neuro deficit, pregnancy, COHb > 25%
- **Cyanide:** lactate > 8, ↑SvO_2 → hydroxocobalamin 5 g
- **TRALI:** ≤ 6 h post-transfusion, fever, hypotension, normal BNP → supportive
- **TACO:** hypertension, overload, ↑BNP → diuretics
- **Fat embolism:** 24–72 h after long-bone fracture; hypoxemia + neuro + petechiae (Gurd)
- **Air embolism:** left lateral + Trendelenburg, 100% O_2, HBO
- **Drowning:** hypoxemia is the key insult; no difference fresh vs salt water
- **Amniotic fluid embolism:** collapse + hypoxemia + DIC during labor

16. High-Yield MCQ Pearls

- [x] Global ARDS 2023: S/F ≤ 315 & HFNC ≥ 30 L/min included
- [x] Prone ≥ 16 h if P/F < 150 → ↓mortality
- [x] iNO ↑oxygenation, no ↓mortality
- [x] Sweep gas → CO_2; blood flow → O_2 (ECMO)
- [x] "Normal" $PaCO_2$ in asthma attack = danger
- [x] COPD NIV when pH < 7.35 & $PaCO_2$ > 45
- [x] Severe CAP: hydrocortisone ↓mortality
- [x] PE with shock → thrombolysis
- [x] Intermediate-high PE: anticoagulate, rescue lysis
- [x] RV failure: keep MAP > PAP, avoid fluids
- [x] Bleomycin lung → avoid high FiO_2
- [x] Pleural pH < 7.2 → drain
- [x] Hemoptysis → bleeding side down + BAE
- [x] CO poisoning: SpO_2 falsely normal

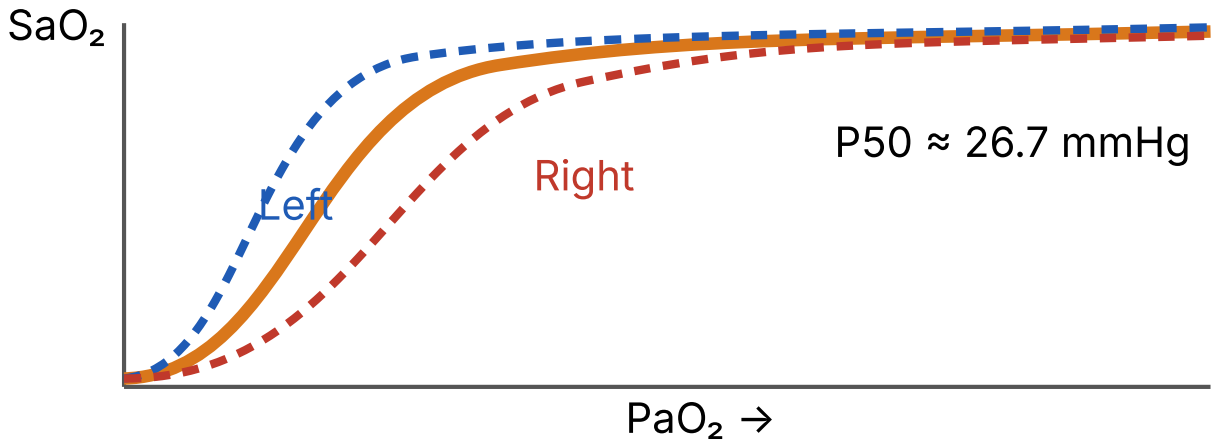
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- **Respiratory system C** = lung + chest wall (in series): $1/C_{rs} = 1/C_L + 1/C_{cw}$
- ↓C_{cw}: obesity, abdominal hypertension, burns eschar, kyphoscoliosis
- **FRC** ↓ supine, obesity, anesthesia, ARDS; ↑ in COPD (hyperinflation)
- **Closing capacity** > FRC (elderly, supine) → airway closure → V/Q mismatch
- **Surfactant** (type II pneumocytes) ↓ surface tension → alveolar stability

2. Control of Breathing

- **Central chemoreceptors** (medulla): respond to CSF H^+ ← $PaCO_2$ – main minute-to-minute drive
- **Peripheral** (carotid > aortic bodies): PaO_2 < 60, ↓pH, ↑$PaCO_2$
- **$PaCO_2 = 0.863 \times VCO_2 / VA$** → CO_2 depends on alveolar ventilation
- Opioids, sedatives, hypothyroid, chronic hypercapnia ↓ CO_2 response slope
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- **Cheyne–Stokes:** HF, stroke (unstable loop gain)
- O_2 in COPD → ↑$PaCO_2$ mainly via **↑V/Q mismatch (loss of HPV) + Haldane effect**, not loss of hypoxic drive

3. Oxygen Transport



- **Right shift (↓affinity):** ↑CO_2, ↑H^+, ↑temp, ↑2,3-DPG
- **Left shift:** CO, MetHb, alkalosis, hypothermia, fetal Hb, stored blood
- **CaO_2** = 1.34 × Hb × SaO_2 + 0.003 × PaO_2 (≈ 20 mL/dL)
- **DO_2** ≈ 1000 mL/min; **VO_2** ≈ 250 mL/min; **O_2ER** ≈ 25%
- **SvO_2** ≈ 65–70%; $ScvO_2$ normally 2–5% higher; ↓ = ↓DO_2 or ↑VO_2; ↑ = sepsis/shunting, cyanide

4. CO_2 & Dead Space

- CO_2 carried: **HCO_3^- ~70%**, carbamino ~20%, dissolved ~10%
- **Haldane effect:** oxygenation of Hb releases CO_2
- **Bohr effect:** ↑CO_2/H^+ releases O_2
- **VD/VT** = ($PaCO_2$ − $PECO_2$)/$PaCO_2$; normal 0.2–0.35; ARDS > 0.6 = ↑mortality
- **Ventilatory ratio** = (VE × $PaCO_2$)/(PBW × 100 × 37.5); > 1 = ↑dead space
- **Causes ↑dead space:** PE, low CO, high PEEP/overdistension, COPD, apparatus (HME)

5. Approach to Hypoxemia

Mechanism	A-a	Response to O_2
↓ PiO_2 (altitude)	Normal	Good
Hypoventilation	Normal	Good
V/Q mismatch	↑	Good
Diffusion limit	↑	Good
Shunt	↑	**Poor (> 30%)**

- **PAO_2** = FiO_2 (Patm − 47) − $PaCO_2$/0.8
- Expected A-a ≈ Age/4 + 4 (room air)
- Intracardiac shunt (PFO with ↑RAP), hepatopulmonary syndrome (orthodeoxia), AVM
- **Low SvO_2** worsens hypoxemia in presence of shunt

6. Acid–Base Essentials

Disorder	Expected compensation
Met acidosis	$PaCO_2 = 1.5 \times HCO_3 + 8 \pm 2$
Met alkalosis	$PaCO_2$ ↑ 0.7 per 1 ↑HCO_3
Acute resp acid.	HCO_3 ↑1 / 10 $PaCO_2$
Chronic resp acid.	HCO_3 ↑3.5 / 10 $PaCO_2$
Acute resp alk.	HCO_3 ↓2 / 10 $PaCO_2$
Chronic resp alk.	HCO_3 ↓5 / 10 $PaCO_2$

- **AG** = Na – (Cl + HCO_3); corrected +2.5 per 1 g/dL ↓albumin
- **Δ-ratio** ($\Delta AG/\Delta HCO_3$): < 1 + NAGMA; 1–2 pure HAGMA; > 2 + met alk
- **Osmolar gap** > 10 → toxic alcohols
- **Stewart:** SIDa normal ≈ 40; ↓SID (hyperchloremia, NaCl 0.9%) → acidosis
- Post-hypercapnic met alkalosis after rapid ventilation of chronic CO_2 retainers

7. Respiratory Monitoring

Pulse oximetry pitfalls

- **CO-Hb:** falsely normal SpO_2 → co-oximetry
- **MetHb:** $SpO_2 \approx 85\%$ regardless of PaO_2 ("saturation gap")
- Poor perfusion, motion, nail polish; **overestimates in dark skin**

Capnography

- Pa-$EtCO_2$ gap normally 2–5 → ↑ with dead space
- Loss of waveform: disconnection / esophageal / dislodged tube
- Shark-fin: bronchospasm; CPR $EtCO_2 < 10$ = poor CPR; abrupt ↑ = ROSC

Esophageal pressure (Pes)

- Transpulmonary P_L = Paw − Pes
- End-inspiratory $P_L < 20–25$ (stress); end-expiratory $P_L \geq 0$ to prevent collapse (obesity)
- **Pmus / ΔPocc** estimate patient effort

8. Lung Ultrasound – BLUE Protocol

Profile	Diagnosis
A-profile + DVT	Pulmonary embolism
A-profile, no DVT, + PLAPS	Pneumonia
A-profile, nothing else	COPD / asthma
B-profile (bilateral B-lines + sliding)	Pulmonary edema
B'-profile (B-lines, no sliding)	Pneumonia
A'-profile + lung point	Pneumothorax

- Lung point = 100% specific for pneumothorax
- Diaphragm thickening fraction < 20–30% → weaning failure risk
- Lung ultrasound score (LUS) tracks aeration / recruitment

9. Oxygen Therapy & HFNC

Device	Flow	Approx. FiO_2
Nasal cannula	1–6 L/min	24–44%
Simple face mask	5–10 L/min	35–55%
Venturi	Set	24–60% (fixed)
Non-rebreather	10–15 L/min	60–90%
HFNC	30–60 L/min	21–100%

HFNC benefits: washout of dead space, small PEEP (~1 cmH_2O / 10 L), precise FiO_2, humidification, ↓work of breathing

- **FLORALI:** HFNC ↓ 90-day mortality vs NIV in hypoxemic ARF (P/F ≤ 300)
- **ROX** = $(SpO_2/FiO_2)/RR$: ≥ 4.88 success; < 3.85 failure at 12 h
- **Targets:** SpO_2 92–96% general; 88–92% hypercapnic risk
- Delayed intubation after HFNC failure → ↑mortality

10. Non-Invasive Ventilation

✔ **Strong evidence:** AECOPD with pH < 7.35 & $PaCO_2$ > 45; cardiogenic pulmonary edema (CPAP or BiPAP); immunocompromised; post-extubation in high-risk / hypercapnic; pre-oxygenation

✘ **Contraindications:** arrest / imminent arrest; coma or unable to protect airway; shock; uncontrolled vomiting / UGIB; facial trauma or surgery; undrained pneumothorax; copious secretions

- **Start:** IPAP 10–12, EPAP 4–5 → titrate IPAP for VT 6–8 mL/kg & ↓$PaCO_2$; EPAP for oxygenation
- Reassess ABG at **1–2 h**; no improvement → intubate
- **HACOR score** > 5 at 1 h predicts NIV failure in hypoxemia
- Helmet NIV: better tolerated, allows higher PEEP

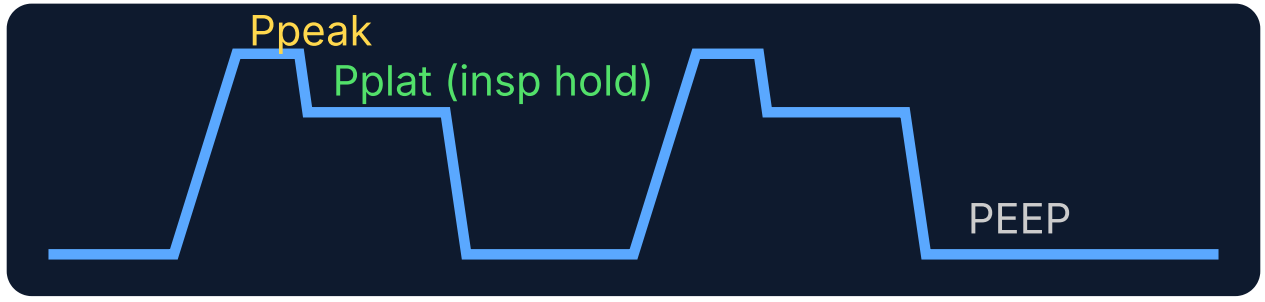
11. Invasive Ventilation – Modes & Setup

Mode	Set	Variable
VC-AC	VT, RR, flow	Pressures
PC-AC	Pinsp, Ti, RR	VT
PRVC	Target VT	Pressure adapts
PSV	PS, PEEP	VT, RR, Ti
SIMV	Mandatory + PS	Mixed
APRV	P-high, T-high, P-low, T-low	VT
NAVA / PAV+	Gain	Proportional to effort

Initial settings: VT 6–8 mL/kg PBW • RR 14–20 • PEEP 5+ • FiO_2 to SpO_2 92–96% • I:E 1:2

- **PBW** male = 50 + 0.91 (height cm − 152.4); female = 45.5 + 0.91 (height cm − 152.4)
- ↑ Oxygenation: ↑FiO_2, ↑PEEP, ↑mean airway pressure
- ↑ CO_2 clearance: ↑RR or VT, ↓dead space (remove HME)

12. Ventilator Troubleshooting



- **↑Ppeak, normal Pplat** → ↑resistance: secretions, bronchospasm, biting, kinked / small ETT
- **↑Ppeak and ↑Pplat** → ↓compliance: pneumothorax, mainstem intubation, atelectasis, edema, ARDS, abdominal hypertension, auto-PEEP

Acute deterioration – "DOPES": Displacement, Obstruction, Pneumothorax, Equipment failure, Stacked breaths (auto-PEEP) → disconnect & bag with 100% O_2

- **Auto-PEEP:** expiratory flow not reaching zero; expiratory hold; ↓RR, ↓VT, ↑expiratory time, bronchodilators
- **Dyssynchrony:** ineffective, double, auto- & reverse triggering, flow starvation, premature / delayed cycling

13. Lung Protection & VILI

Targets: VT 6 mL/kg PBW (4–8) • Pplat ≤ 30 • ΔP (Pplat − PEEP) ≤ 15 • pH ≥ 7.20–7.25

- **ARDSNet (ARMA):** 6 vs 12 mL/kg → ↓ mortality (31% vs 40%)
- **VILI:** barotrauma, volutrauma, atelectrauma, biotrauma, ergotrauma
- **Stress** = P_L; **strain** = VT/FRC (> 2 injurious)
- **Mechanical power** (energy/min) > ~17 J/min associated with mortality
- **P-SILI:** vigorous spontaneous efforts → large P_L swings, pendelluft
- Protective VT also matters in non-ARDS patients (aim 6–8)
- **Recruitment maneuvers:** not routine (ART: ↑mortality)

14. Liberation & Tracheostomy

- **Readiness:** cause improving, FiO_2 ≤ 0.4–0.5, PEEP ≤ 5–8, P/F > 150–200, minimal vasopressors, follows commands
- **SBT:** PS 5–8 / PEEP 0–5 or T-piece, 30–120 min
- **Fail if:** RR > 35, SpO_2 < 90%, HR > 140 or ↑20%, SBP > 180 or < 90, anxiety, diaphoresis, paradoxical breathing
- **RSBI** = RR/VT (L) < 105
- **Daily SAT + SBT** (ABC trial) ↓ ventilator days & mortality
- **Post-extubation:** HFNC (low risk); NIV ± HFNC (high risk, hypercapnic, obese)
- **Cuff-leak** < 110 mL or < 12–24% → methylprednisolone ≥ 4 h before
- **Weaning failure causes:** cardiac (weaning-induced pulmonary edema), weakness, ↑load, delirium, metabolic
- **Tracheostomy:** early (≤ day 4) vs late (≥ day 10) – no mortality benefit (TracMan)

15. Airway Management

- **Physiologically difficult airway:** hypoxemia, hypotension, severe metabolic acidosis, RV failure
- **MACOCHA** ≥ 3 predicts difficult ICU intubation
- **Pre-oxygenation:** NIV better than face mask (PREOXI); HFNC for apneic oxygenation
- **Video laryngoscopy** ↑ first-pass success (DEVICE)
- **Induction:** ketamine / etomidate – reduce dose in shock; push-dose vasopressor ready
- **Rocuronium** 1.2 mg/kg ≈ succinylcholine 1.5 mg/kg
- **Sux CI:** hyperkalemia, burns / crush > 24–48 h, denervation, myopathy, MH
- Confirm with **waveform capnography**

CICO → scalpel–bougie–tube cricothyroidotomy

16. High-Yield MCQ Pearls

- ✔ Hypoventilation & altitude: normal A-a gradient
- ✔ Shunt > 30%: PaO_2 barely rises with FiO_2
- ✔ CO poisoning: normal SpO_2 – use co-oximetry
- ✔ MetHb: SpO_2 fixed ≈ 85%
- ✔ ↑Ppeak + normal Pplat = resistance problem
- ✔ Auto-PEEP + hypotension → disconnect
- ✔ ΔP is the strongest predictor of ARDS survival
- ✔ HFNC beat NIV in hypoxemic ARF (FLORALI)
- ✔ P0.1 > 4 cmH_2O: excessive drive
- ✔ Lung point = pneumothorax (100% specific)
- ✔ Pair SAT + SBT daily
- ✔ Early tracheostomy: no mortality benefit

1. ARDS – Definition & Pathology

Global definition (2023): onset ≤ 1 week; bilateral opacities (CXR / CT / ultrasound); not fully explained by cardiac failure / overload

Category	Oxygenation criterion
Intubated	P/F ≤ 300 or S/F ≤ 315, PEEP ≥ 5 Mild 200–300 • Mod 100–200 • Severe ≤ 100
Non-intubated	HFNC ≥ 30 L/min or NIV/CPAP ≥ 5; P/F ≤ 300 or S/F ≤ 315
Resource-limited	S/F ≤ 315 (no PEEP / flow requirement)

- **Direct:** pneumonia, aspiration, contusion, inhalation, drowning
- **Indirect:** sepsis (most common), pancreatitis, trauma, transfusion, burns
- **Pathology:** diffuse alveolar damage – exudative (hyaline membranes) → proliferative → fibrotic
- S/F valid only when SpO_2 ≤ 97%

2. ARDS – Management Ladder



- **Treat cause**, conservative fluids after shock resolves (FACTT)
- **VT 6 mL/kg PBW**, Pplat ≤ 30, ΔP ≤ 15 (ARMA)
- **Higher PEEP** in moderate–severe (PEEP/FiO_2 table, P_L-guided or best compliance)
- **Prone ≥ 16 h/day** if P/F < 150 on FiO_2 ≥ 0.6 – ↓mortality (PROSEVA)
- **NMB** 48 h: no routine benefit (ROSE); use for severe dyssynchrony / refractory hypoxemia
- **Inhaled vasodilators:** improve oxygenation, no mortality benefit (rescue only)
- **Steroids:** dexamethasone early in moderate–severe (DEXA-ARDS); COVID & severe CAP benefit
- **Avoid:** HFOV (OSCILLATE), routine recruitment (ART), β-agonists

3. ECMO

	VV-ECMO	VA-ECMO
Supports	Gas exchange	Heart + lungs
Indication	Refractory ARDS, bridge to transplant	Cardiogenic shock, ECPR, massive PE
Cannulas	Femoral → IJ	Femoral vein → femoral artery

EOLIA criteria: P/F < 50 for > 3 h, or < 80 for > 6 h, or pH < 7.25 with $PaCO_2$ ≥ 60 for > 6 h (despite optimal care)

- **Relative CI:** MV > 7 days at high settings, irreversible disease, contraindication to anticoagulation, severe brain injury
- **Ultra-protective** ventilation on ECMO (rest settings, Pplat ≤ 25)
- **Oxygenation** ↑ with blood flow; **CO_2 removal** ↑ with sweep gas flow
- **Harlequin (north–south) syndrome:** peripheral VA + poor lung function → hypoxic upper body (monitor right radial ABG)
- VV recirculation ↓ efficiency; $ECCO_2R$ alone: no benefit (REST)

4. Life-Threatening Asthma

Life-threatening: SpO_2 < 92%, PEF < 33%, silent chest, cyanosis, exhaustion, confusion, arrhythmia, hypotension, **"normal" $PaCO_2$**
Near-fatal: ↑$PaCO_2$ or need for ventilation

- **Rx:** O_2 to SpO_2 93–95%, nebulized salbutamol (back-to-back) + ipratropium, hydrocortisone 100 mg IV / prednisolone 40–50 mg, **$MgSO_4$ 2 g IV over 20 min**
- **Refractory:** IV salbutamol / adrenaline, ketamine, heliox, inhaled anesthetics
- **Salbutamol toxicity:** lactic acidosis, hypokalemia, tachycardia
- **Intubate with ketamine**, large ETT
- **Ventilate:** low RR (8–12), VT 6–8, high inspiratory flow, I:E ≥ 1:4, permissive hypercapnia, Pplat < 30, minimal external PEEP
- Peri-intubation arrest → disconnect, compress chest, exclude tension pneumothorax

5. AECOPD

- **Triggers:** viral > bacterial infection, pollution, PE, HF, non-adherence
- **Controlled O_2:** SpO_2 88–92% (Venturi)
- **Bronchodilators:** SABA ± SAMA nebulized
- **Steroids:** prednisolone 40 mg × 5 days (REDUCE)
- **Antibiotics** if ↑sputum purulence + ↑volume or ↑dyspnea (Anthonisen) or ventilated

NIV: pH < 7.35 & $PaCO_2$ > 45 → ↓intubation & mortality; pH < 7.25 → NIV with low threshold for intubation

- **Ventilation:** long expiratory time, monitor auto-PEEP, target baseline $PaCO_2$ (avoid post-hypercapnic alkalosis)
- **Mimics:** PE, pneumothorax, HF, pneumonia

6. Severe Community-Acquired Pneumonia

ATS/IDSA severe CAP: 1 major (septic shock on vasopressors or MV) or ≥ 3 minor: RR ≥ 30, P/F ≤ 250, multilobar, confusion, BUN ≥ 20, WBC < 4000, plt < 100k, T < 36 °C, hypotension needing fluids

- **Empiric:** β-lactam + macrolide (or + respiratory FQ)
- Add **MRSA** / **Pseudomonas** cover only if risk factors (prior isolation, recent hospitalization + IV antibiotics)
- **Hydrocortisone** 200 mg/day in severe CAP → ↓mortality (CAPE COD)
- **Duration:** ≥ 5 days & clinically stable

Organism	Clue
Legionella	Hyponatremia, diarrhea, ↑LFT, urine antigen
S. aureus (PVL)	Post-influenza, necrotizing, hemoptysis
Klebsiella	Alcohol, upper lobe, cavitation
Pneumococcus	Most common; urine antigen

7. HAP / VAP

- **HAP:** ≥ 48 h after admission; **VAP:** ≥ 48 h after intubation
- **Diagnosis:** new infiltrate + fever, purulent secretions, ↑WBC, ↓oxygenation; quantitative ETA / BAL
- **Empiric:** anti-pseudomonal β-lactam; add MRSA cover (vancomycin / linezolid) if risk
- **MDR risk:** IV antibiotics in prior 90 days, septic shock, ARDS before VAP, ≥ 5 days hospital, acute RRT
- Double Gram-negative cover if MDR risk + high local resistance
- **Duration:** 7 days (de-escalate by culture)

Prevention bundle: head-up 30–45°, daily SAT/SBT, subglottic secretion drainage, minimize sedation, early mobility, cuff pressure 20–30 cmH_2O

8. Viral & Opportunistic Infections

- **Influenza:** oseltamivir for all severe cases, even > 48 h; watch S. aureus superinfection, aspergillosis
- **COVID-19:** dexamethasone 6 mg × 10 d if O_2-requiring; + IL-6 inhibitor or baricitinib if progressing; awake prone positioning
- **PCP:** CD4 < 200, ↑LDH, ↑β-D-glucan, diffuse GGO, pneumothorax; TMP-SMX + steroids if PaO_2 < 70 or A-a ≥ 35
- **Invasive aspergillosis:** neutropenia, halo / air-crescent sign, BAL galactomannan → voriconazole
- **Mucormycosis:** diabetic ketoacidosis, reverse halo → liposomal amphotericin + debridement
- **CMV:** transplant → ganciclovir
- **Nocardia:** cavities + brain abscess → TMP-SMX
- **TB:** airborne isolation; RIPE (INH neuropathy – give pyridoxine; ethambutol optic neuritis; rifampicin interactions)

9. Pulmonary Embolism

Risk	Features	Management
High	SBP < 90 ≥ 15 min, vasopressors, arrest	Systemic thrombolysis (alteplase 100 mg / 2 h; arrest 50 mg bolus); embolectomy / CDT / VA-ECMO if contraindicated
Int-high	RV dysfunction + ↑troponin	Anticoagulation, monitor; rescue reperfusion if deteriorates (PEITHO: ↑bleeding)
Int-low	sPESI ≥ 1, one marker	Anticoagulation
Low	sPESI 0	DOAC, early discharge

- **Pre-test:** Wells / Geneva; PERC (low risk); age-adjusted D-dimer (age × 10 µg/L)
- **Imaging:** CTPA; V/Q in pregnancy / renal failure; unstable → bedside echo (RV dilatation, McConnell sign)
- **RV failure:** small fluid boluses only, norepinephrine, avoid intubation if possible (collapse)
- **Anticoagulation:** UFH if thrombolysis likely / renal failure; otherwise LMWH / DOAC

10. Pulmonary Hypertension & RV Failure

- **PH (2022):** mPAP > 20 mmHg on RHC
- **Pre-capillary:** PAWP ≤ 15 + PVR > 2 WU; **post-capillary:** PAWP > 15
- **Groups:** 1 PAH • 2 left heart disease (most common) • 3 lung disease / hypoxia • 4 CTEPH • 5 unclear / multifactorial
- **PAH drugs:** PDE5i, ERA, riociguat, prostacyclin analogues

ICU RV failure:

• Preload: avoid overload, diuresis / CRRT if congested

• Afterload: correct hypoxia, hypercapnia, acidosis; iNO / inhaled epoprostenol

• Contractility: dobutamine, milrinone, levosimendan

• Perfusion: keep MAP > PAP (norepinephrine, vasopressin)

• Sinus rhythm; avoid high PEEP / Pplat; intubation = high-risk event

11. ILD in the ICU

Acute exacerbation of IPF: new bilateral GGO within 1 month, not explained by HF / overload; mortality > 50%; MV outcomes very poor → discuss goals / transplant candidacy

- **Rx (weak evidence):** high-dose steroids, broad antibiotics; cyclophosphamide harmful (EXAFIP)
- **Rapidly progressive ILD:** anti-MDA5 dermatomyositis, anti-synthetase syndrome, vasculitis
- **Drug-induced:** amiodarone, methotrexate, nitrofurantoin, checkpoint inhibitors, **bleomycin (avoid high FiO_2)**
- **Hypersensitivity pneumonitis:** exposure, mosaic attenuation, BAL lymphocytosis
- **Sarcoidosis:** non-caseating granulomas, hilar nodes, hypercalcemia
- **Organizing pneumonia:** migratory consolidations, reverse halo, steroid-responsive
- **AEP (acute eosinophilic pneumonia):** new smoker, BAL eosinophils > 25%, dramatic steroid response

12. Pleural Disease

Light's criteria (exudate if ANY)

- Pleural / serum protein > 0.5
- Pleural / serum LDH > 0.6
- Pleural LDH > 2/3 ULN serum LDH
- Diuresed HF misclassified as exudate → serum–pleural albumin gradient > 1.2 g/dL = transudate

Finding	Meaning
pH < 7.2 / glucose < 3.3 mmol/L	Complicated parapneumonic → drain
Frank pus	Empyema → drain
TG > 110 mg/dL	Chylothorax
Hct ratio > 0.5	Hemothorax
ADA > 40 U/L	TB
Amylase ↑	Pancreatitis, esophageal rupture

- **Empyema:** antibiotics + drain + intrapleural tPA 10 mg / DNase 5 mg BD × 3 d (MIST-2); VATS if fails
- **Re-expansion edema:** drain ≤ 1.5 L at once

13. Pneumothorax & Air Leak

Tension: clinical diagnosis – hypotension, ↑Ppeak/Pplat, desaturation, absent breath sounds → immediate decompression (4th–5th ICS anterior axillary line; or 2nd ICS MCL), then chest drain

- **Ventilated patients:** any pneumothorax → chest drain
- **US:** absent sliding, no B-lines, barcode sign (M-mode), lung point
- **Bronchopleural fistula:** persistent air leak; minimize Pplat / PEEP / VT / Ti, lowest suction; endobronchial valves or surgery; independent lung ventilation (severe)
- **Hemothorax:** large-bore drain; thoracotomy if > 1500 mL initially or > 200 mL/h for 2–4 h

14. Massive Hemoptysis & DAH

- **Massive:** causing airway compromise / hypoxemia (volume definitions 100–600 mL/24 h)
- **Source:** bronchial arteries ~90%
- **Causes:** bronchiectasis, TB, aspergilloma, malignancy, CF, vasculitis

Rx: bleeding side down • secure airway (ETT ≥ 8 for bronchoscopy; selective intubation of non-bleeding lung / bronchial blocker) • reverse coagulopathy • TXA • CT angiography → **bronchial artery embolization** • surgery if fails

- **DAH:** ↓Hb, diffuse infiltrates, progressively bloodier BAL, hemosiderin-laden macrophages, ↑DLCO
- **DAH causes:** GPA, MPA, anti-GBM, SLE, HSCT, coagulopathy, cocaine
- **DAH Rx:** pulse methylprednisolone + cyclophosphamide / rituximab; plasma exchange in anti-GBM

15. Other Respiratory Emergencies

- **Smoke inhalation:** early intubation if stridor, facial burns, soot, hoarseness
- **CO poisoning:** normal SpO_2; 100% O_2 (t½ 320 → 74 min); HBO if LOC, neuro deficit, pregnancy, COHb > 25%
- **Cyanide:** lactate > 8, ↑SvO_2 → hydroxocobalamin 5 g
- **TRALI:** ≤ 6 h post-transfusion, fever, hypotension, normal BNP → supportive
- **TACO:** hypertension, overload, ↑BNP → diuretics
- **Fat embolism:** 24–72 h after long-bone fracture; hypoxemia + neuro + petechiae (Gurd)
- **Air embolism:** left lateral + Trendelenburg, 100% O_2, HBO
- **Drowning:** hypoxemia is the key insult; no difference fresh vs salt water
- **Amniotic fluid embolism:** collapse + hypoxemia + DIC during labor

16. High-Yield MCQ Pearls

- ✔ Global ARDS 2023: S/F ≤ 315 & HFNC ≥ 30 L/min included
- ✔ Prone ≥ 16 h if P/F < 150 → ↓mortality
- ✔ iNO ↑oxygenation, no ↓mortality
- ✔ Sweep gas → CO_2; blood flow → O_2 (ECMO)
- ✔ "Normal" $PaCO_2$ in asthma attack = danger
- ✔ COPD NIV when pH < 7.35 & $PaCO_2$ > 45
- ✔ Severe CAP: hydrocortisone ↓mortality
- ✔ PE with shock → thrombolysis
- ✔ Intermediate-high PE: anticoagulate, rescue lysis
- ✔ RV failure: keep MAP > PAP, avoid fluids
- ✔ Bleomycin lung → avoid high FiO_2
- ✔ Pleural pH < 7.2 → drain
- ✔ Hemoptysis → bleeding side down + BAE
- ✔ CO poisoning: SpO_2 falsely normal